

Stability of Local Feature Explanations for Graph Neural Networks under Controlled Perturbations

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Abstract

Graph neural network explanations can provide insight into the features associated with individual predictions, but the stability of these explanations under changes to the input remains an important consideration. This study investigates the stability of local GraphLIME feature explanations for a frozen graph convolutional network under controlled feature perturbations. Experiments were conducted on the Cora citation network using a validation-selected two-layer graph convolutional network and a fixed cohort of 97 target nodes with valid baseline GraphLIME explanations. Active feature entries within each target node's two-hop explanation neighbourhood were randomly masked at rates of 1%, 5%, 10%, and 20%, with ten perturbation repetitions per node and rate, yielding 3,880 perturbation observations. Explanation stability was measured using top-10 Jaccard similarity, while prediction agreement and explanation availability were evaluated separately. Mean explanation overlap decreased from 0.9015 at 1% masking to 0.4541 at 20% masking, whereas prediction stability remained comparatively high, decreasing from 0.9969 to 0.9588. When analysis was restricted to perturbations preserving the baseline predicted class, explanation overlap similarly decreased from 0.9014 to 0.4560. Node-level changes in prediction and explanation stability exhibited little descriptive association, while complete explanations remained available for more than 98% of perturbation observations at every masking rate. These findings demonstrate that, within the fixed Cora, GCN, and GraphLIME configuration studied here, preservation of a categorical prediction does not imply preservation of its local feature explanation. The results motivate explicit evaluation of explanation stability as a property distinct from predictive stability and explanation availability.

Keywords: graph neural networks; explainable artificial intelligence; GraphLIME; explanation stability; robustness

Introduction

Graph-structured data arise naturally in domains where observations are connected through relational information, including citation networks, social systems, molecular structures, and knowledge graphs. Graph neural networks (GNNs) provide a general framework for learning from such data by combining node attributes with information propagated through graph structure. In particular, graph convolutional networks (GCNs) extend neural representation learning to graph domains through neighbourhood aggregation and have become a foundational architecture for node-level prediction [1]. Despite their predictive capabilities, however, the nonlinear interaction between node attributes, graph topology, and repeated message passing can make the basis of individual GNN predictions difficult to interpret. This has motivated substantial research into methods for explaining trained GNNs [2,3].

Post-hoc explanation methods address this problem by analysing a trained model without requiring its predictive architecture to be redesigned. Depending on the method, an explanation may identify influential nodes, edges, subgraphs, or node features associated with a particular prediction. Methods such as GNNExplainer, PGExplainer, SubgraphX, and GraphLIME illustrate the range of approaches developed for this purpose [4-7]. As the number and diversity of GNN explanation methods have increased, evaluating the quality and reliability of their outputs has become an important problem in its own right. In particular, the absence of reliable ground-truth explanations for many real graph datasets makes explanation evaluation fundamentally different from evaluating predictive accuracy [8].

One aspect of this evaluation problem is **explanation stability**: whether an explanation remains similar when the input undergoes controlled changes. Stability is distinct from predictive stability. A perturbation may leave a model's predicted class unchanged while altering the explanation produced for that prediction. Consequently, observing that a GNN continues to make the same categorical decision does not, by itself, establish that its post-hoc explanation remains unchanged. This distinction is especially relevant when explanations are used to characterize local model behaviour, because users may otherwise interpret stable predictions as evidence that the explanatory rationale is also stable.

Recent research has provided direct evidence that predictive and explanatory robustness can diverge for GNNs. Li et al. showed that limited adversarial modifications to graph structure can cause substantial changes in explanations while preserving correct GNN predictions [9]. More broadly, systematic evaluation frameworks and recent surveys emphasize that explanation reliability is multi-dimensional and cannot be inferred solely from model accuracy or a single explanation-quality criterion [2, 3, 8, 10]. These developments motivate controlled studies in which predictive behaviour and explanatory behaviour are measured separately under explicitly defined perturbations.

This study focuses on **GraphLIME**, a local post-hoc explanation method designed to identify informative node features for GNN predictions [7]. For a target node, GraphLIME constructs a local dataset from its graph neighbourhood and the outputs of the trained GNN, then uses kernel-based nonlinear feature selection through an HSIC-Lasso objective to obtain a sparse set of explanatory features [7, 11]. This formulation makes GraphLIME particularly suitable for studying feature-level explanation stability: after a baseline explanation has been obtained, node-feature information can be perturbed while the graph structure and trained GNN remain fixed, allowing changes in the resulting feature explanation to be measured directly.

The central question investigated in this work is therefore:

How stable are GraphLIME local feature explanations for a frozen graph neural network under controlled perturbations of the input node features?

The study addresses this question using the Cora citation network and a frozen two-layer GCN. A predetermined cohort of correctly classified test nodes is explained using a fixed GraphLIME configuration. Active feature entries within each target node's two-hop explanation neighbourhood are then randomly masked at rates of 1%, 5%, 10%, and 20%, with repeated perturbations at each severity. The GNN is not retrained after perturbation. Explanation stability is quantified through top-10 Jaccard overlap between baseline and perturbed GraphLIME feature sets, while prediction agreement and explanation availability are measured separately.

The experimental design is intended to distinguish three related but conceptually different outcomes. First, **prediction stability** asks whether the perturbed input retains the baseline predicted class. Second, **explanation stability** asks whether the features selected by GraphLIME remain similar to those in the baseline explanation. Third, **explanation availability** asks whether the explanation procedure continues to return a complete top-10 explanation under the

predefined validity criteria. Keeping these quantities separate prevents explanation disagreement from being conflated with either prediction change or explanation-generation failure.

Particular attention is given to perturbations for which the predicted class remains unchanged. If explanation overlap decreases even within this prediction-preserving subset, then explanation variation cannot be attributed solely to changes in the GCN's categorical decision. The study therefore evaluates explanation stability both unconditionally and conditional on prediction agreement. It additionally examines node-level coupling between changes in prediction stability and explanation stability across perturbation severity.

The contribution of this work is a **controlled empirical robustness analysis of local GraphLIME feature explanations**. Rather than proposing a new explainer, the study evaluates an existing explanation method under a fixed and reproducible perturbation protocol. Specifically, it (i) quantifies how top-10 GraphLIME explanation overlap changes as feature-masking severity increases; (ii) tests whether the same pattern persists when the GCN's predicted class is preserved; (iii) evaluates whether changes in prediction stability and explanation stability are coupled across target nodes; and (iv) distinguishes explanation instability from explanation unavailability. This positioning complements existing work on GNN explanation evaluation and robustness while addressing a feature-perturbation setting that differs from adversarial structural attacks [8-10].

The results show that these stability dimensions behave differently within the fixed experimental setting. As masking severity increases from 1% to 20%, mean top-10 GraphLIME explanation overlap decreases substantially, while prediction agreement remains comparatively high. The decline in explanation overlap persists when analysis is restricted to perturbations preserving the predicted class, and node-level changes in predictive and explanatory stability exhibit little descriptive association. At the same time, complete explanations remain available for more than 98% of perturbation observations at every masking rate. These findings demonstrate, for the Cora/GCN/GraphLIME configuration studied here, that stable categorical predictions and high explanation availability can coexist with substantial changes in the selected local explanatory features.

The remainder of this report is organized as follows. Section 2 reviews GNN explainability, GraphLIME, and prior work on explanation evaluation and stability. Section 3 describes the dataset, frozen GCN, GraphLIME implementation, controlled feature-perturbation protocol, stability measures, and statistical analysis. Section 4 presents the experimental results. Section 5

discusses their interpretation in relation to explanation robustness, and Section 6 describes the limitations of the study. Section 7 concludes the report and summarizes directions for further investigation.

Related Work

Explainability for Graph Neural Networks

Graph neural networks (GNNs) learn representations by combining node attributes with information propagated through graph structure, enabling effective modelling of relational data but also making individual predictions difficult to interpret. Consequently, a substantial body of work has developed post-hoc explanation methods for identifying the nodes, edges, subgraphs, or node features that contribute to GNN predictions [2,3]. Existing methods span several methodological families, including gradient-based attribution, perturbation-based explanation, surrogate modelling, probabilistic approaches, and optimization-based methods. They also differ in explanatory scope: some methods identify influential graph structure, whereas others estimate the contribution of node attributes or construct local explanations around a particular prediction.

Early and representative methods illustrate this methodological diversity. GNNExplainer formulates explanation generation as an optimization problem for identifying compact subgraphs and feature subsets that are informative about a GNN prediction [4]. PGExplainer instead learns a parameterized explanation model that can generate explanations for multiple instances [5], while SubgraphX searches for explanatory subgraphs using a Shapley-value-inspired formulation [6]. Other approaches derive explanations from gradients, perturbations, probabilistic graphical models, or local surrogate models. Surveys of GNN explainability emphasize that these methods differ not only in how explanations are constructed but also in the explanatory objects they produce and the assumptions underlying their interpretation [2,3].

The diversity of explanation mechanisms has made evaluation a central problem in GNN explainability. GraphFramEx proposed a systematic framework for evaluating input-level GNN explainers and demonstrated that no single method dominates across all evaluation dimensions [10]. GraphXAI similarly provides datasets, explanation interfaces, and evaluation metrics for systematic benchmarking, including synthetic graph generation with known ground-truth explanations [8]. Such work highlights an important distinction between predictive performance and explanation quality: a GNN can make an

accurate prediction without this alone establishing that a post-hoc explanation is correct, faithful, robust, or otherwise reliable. More recent surveys continue to identify evaluation methodology and the technical limitations of post-hoc explanation techniques as important challenges for explainable graph learning [3].

GraphLIME and Local Feature Explanations

GraphLIME was introduced by Huang et al. as a local post-hoc explanation method for node-level GNN predictions [7]. Rather than explaining a prediction primarily through an influential subgraph, GraphLIME focuses on identifying informative **node features** in the local neighbourhood of a target node. For a node to be explained, the method constructs a local dataset from its N -hop neighbourhood and the corresponding outputs of the trained GNN. It then applies a nonlinear feature-selection procedure based on the Hilbert–Schmidt Independence Criterion Lasso (HSIC-Lasso) to identify features associated with variation in the model outputs [7, 11].

This formulation distinguishes GraphLIME from linear local-surrogate approaches. By representing individual features and model outputs through kernels, GraphLIME can capture nonlinear statistical relationships within the local neighbourhood [7]. The resulting sparse feature coefficients provide a ranking from which a compact set of locally informative features can be selected. Because the procedure operates after model training, it can be applied to a frozen GNN without modifying its learned parameters.

GraphLIME is particularly relevant when the explanatory objective concerns **which node attributes are locally associated with a prediction**, rather than solely which edges or neighbouring nodes are important. Contemporary surveys therefore position GraphLIME among feature-oriented or nonlinear local post-hoc GNN explanation methods [2,3]. However, the reliability of a local explanation involves more than obtaining a sparse feature set for a single unperturbed input. If explanations are intended to characterize local model behaviour, it is also relevant to ask whether the selected features remain similar when the explained input undergoes controlled changes.

The original GraphLIME work primarily evaluates the usefulness and descriptive quality of its generated feature explanations [7]. Robustness under controlled perturbations is therefore a distinct evaluation question from the original objective of constructing locally informative feature explanations. This distinction motivates examining GraphLIME not by changing its explanatory objective, but by evaluating the stability of the resulting selected feature sets when the

underlying GNN is held fixed and its input features are perturbed.

Evaluation and Stability of GNN Explanations

Evaluation of GNN explanations is challenging because real-world graph datasets generally do not provide definitive ground-truth explanations. GraphXAI addresses this problem partly through synthetic graph generation with known explanatory structure, enabling explanation correctness to be evaluated under controlled conditions [8]. GraphFramEx approaches evaluation from a complementary perspective by assessing explanation methods according to criteria related to the sufficiency and necessity of the information selected by an explainer [10]. Together, these frameworks demonstrate that explanation evaluation cannot be reduced to the predictive accuracy of the underlying GNN.

Stability provides another perspective on explanation reliability. In general, stability concerns whether similar or controlled variants of an input receive similar explanations when large explanatory changes are not expected solely from the perturbation. Recent surveys of GNN explainability increasingly discuss robustness and stability as relevant evaluation dimensions alongside more established criteria such as fidelity and sparsity [2,3]. This is particularly important for post-hoc explanations because the prediction and the explanation are produced by distinct computational processes: an input modification may leave the model's final decision unchanged while altering the explanation generated for that decision.

Direct evidence of this separation has begun to emerge in graph explainability research. Li et al. studied GNN explainers under adversarial structural perturbations and showed that small modifications to graph structure can produce substantially different explanations while the underlying GNN continues to make correct predictions [9]. Their experiments establish that preservation of predictive behaviour does not, by itself, guarantee robustness of a GNN explanation. The study focuses on adversarial changes to graph structure and evaluates multiple GNN explainers, making its perturbation model different from controlled non-adversarial feature masking. Nevertheless, it demonstrates the broader importance of evaluating explanatory behaviour separately from predictive behaviour.

This distinction is particularly relevant for local feature explanations. A GNN prediction is ultimately determined by its output decision rule, whereas an explainer such as GraphLIME estimates local feature importance from relationships among neighbourhood features and model outputs. Consequently, preservation of the predicted class does not mathematically require preservation of the explana-

tion produced by the post-hoc procedure. Prediction stability and explanation stability should therefore be measured as related but distinct empirical properties rather than treating one as a proxy for the other.

Positioning of the Present Study

The present study builds on these strands of work by focusing specifically on the **stability of local GraphLIME feature explanations under controlled feature perturbations**. Its purpose is not to introduce a new GNN explainer or to benchmark GraphLIME against competing explanation algorithms. Instead, it treats the explanation returned by a fixed GraphLIME configuration as an object whose behaviour can itself be measured under increasing perturbation severity.

The experimental design differs from adversarial robustness studies such as Li et al. [9] in two important respects. First, the perturbations modify active node-feature entries while leaving graph connectivity unchanged. Second, perturbations are generated according to predetermined masking rates rather than optimized adversarially to maximize explanation disagreement. The resulting experiment therefore addresses a narrower question: how does a local feature explanation behave when increasing amounts of feature information are removed under a controlled, reproducible protocol?

The study also separates three quantities that can otherwise be conflated. **Prediction stability** measures whether the frozen GCN retains its baseline predicted class. **Explanation stability** measures whether the top- K GraphLIME feature set remains similar to the corresponding baseline explanation. **Explanation availability** records whether a complete explanation can be obtained under the predefined validity criteria. Analysing these quantities separately makes it possible to distinguish a changed prediction, a changed but successfully generated explanation, and a failure to obtain a complete explanation.

A central component of this design is the prediction-conditioned analysis. Explanation overlap is evaluated not only across all valid perturbation observations but also specifically among observations for which the predicted class remains unchanged. This provides a direct empirical test of whether explanatory variation persists when categorical predictive behaviour is preserved. The node-level coupling analysis further examines whether changes in prediction stability track changes in explanation stability across target nodes.

Accordingly, the contribution of the present study is best characterized as a **controlled empirical robustness analysis of**

GraphLIME feature explanations, rather than as a new explanation method or a general benchmark of GNN explainability. Within a fixed Cora graph, frozen GCN, predetermined target-node cohort, and fixed GraphLIME configuration, the study evaluates how local feature explanations change under increasing feature masking and whether those changes are coupled to changes in predictive behaviour. This positioning connects the original local-feature explanation objective of GraphLIME [7] with the broader contemporary concern that explanation reliability should be evaluated independently from predictive performance [8-10].

Methodology

Problem Formulation

Let $G = (V, E, X)$ denote an attributed graph, where V is the set of nodes, E is the set of edges, and $X \in \mathbb{R}^{|V| \times d}$ is the node-feature matrix with d input features. Let f denote a trained graph neural network (GNN) that is held fixed throughout the study. For a target node $v \in V$, the model produces a class-probability vector

$$p_0(v) = f(G, v),$$

with predicted class

$$\hat{y}_0(v) = \arg \max_c [p_0(v)]_c. \quad (1)$$

A local post-hoc explanation method associates the prediction for v with a set of influential input features. In this study, GraphLIME [7] is used to obtain a ranked local feature explanation. Let

$$E_0^{(K)}(v) \subseteq \{1, \dots, d\}$$

denote the set of the K highest-ranked features in the baseline GraphLIME explanation for node v , where the experimental protocol fixes $K = 10$.

The central question is whether this local explanation remains stable when the input features are subjected to small, controlled perturbations while the trained GNN and graph structure remain unchanged. Let $X_{r,s}^{(v)}$ denote the feature matrix obtained by applying the prescribed feature-masking perturbation for target node v at masking rate r and perturbation seed s . The corresponding perturbed model output and predicted class are

$$p_{r,s}(v) = f(V, E, X_{r,s}^{(v)}, v), \hat{y}_{r,s}(v) = \arg \max_c [p_{r,s}(v)]_c. \quad (2)$$

Applying GraphLIME to the perturbed input produces a corresponding top- K explanation $E_{r,s}^{(K)}(v)$, when a valid explanation is available.

This formulation distinguishes two conceptually different notions of stability. **Prediction stability** concerns whether the model retains its predicted class after perturbation and is represented at the observation level by

$$P_{r,s}(v) = \mathbb{I} [\hat{y}_{r,s}(v) = \hat{y}_0(v)], \quad (3)$$

where $\mathbb{I}[\cdot]$ is the indicator function. **Explanation stability**, by contrast, concerns whether the features identified as locally influential remain consistent. For observations with valid baseline and perturbed explanations, explanation stability is quantified using top- K Jaccard similarity,

$$J_{r,s}^{(K)}(v) = \frac{|E_0^{(K)}(v) \cap E_{r,s}^{(K)}(v)|}{|E_0^{(K)}(v) \cup E_{r,s}^{(K)}(v)|}. \quad (4)$$

The quantity $J_{r,s}^{(K)}(v)$ lies in $[0, 1]$, with 1 indicating identical top- K feature sets and 0 indicating no overlap. If a valid perturbed explanation cannot be obtained, the Jaccard similarity is treated as unavailable rather than assigned a value of zero; explanation availability is analysed separately.

The distinction between Eqs. (3) and (4) is fundamental to the study. Preservation of the predicted class does not imply preservation of the local explanation. In particular, it is possible to observe

$$\hat{y}_{r,s}(v) = \hat{y}_0(v) \text{ while } E_{r,s}^{(K)}(v) \neq E_0^{(K)}(v). \quad (5)$$

Consequently, evaluating prediction robustness alone does not establish that the explanatory representation of a prediction is robust. The primary objective of this study is therefore to characterize how GraphLIME explanation stability changes as controlled feature perturbations increase in severity, with particular attention to observations for which the GNN's predicted class remains unchanged. Prediction stability and explanation availability are analysed alongside explanation overlap to determine whether these aspects of robustness exhibit corresponding or divergent behaviour.

This formulation defines the quantities evaluated throughout the subsequent experiments. The construction of the GNN, the GraphLIME explanation procedure, the controlled perturbation operator, the stability metrics, and the statistical analysis are specified in the following subsections.

Dataset

Experiments were conducted on the **Cora citation-network dataset** [12], represented as a single attributed graph in which nodes correspond to scientific publications and edges represent citation relationships. Each node is associated with a sparse feature vector describing the publication and a categorical class label. The dataset was loaded using the Planetoid dataset interface provided by PyTorch Geometric [13], with `NormalizeFeatures` applied during loading. The predefined training, validation, and test masks supplied with the dataset representation were retained throughout the study.

The baseline graph convolutional network was trained using the predefined training split, while the validation split was used for model selection. The test split was not used to select the model. After the selected model parameters had been restored and frozen, test-set predictions were computed and the correctly classified test nodes were identified. A deterministic random permutation generated with selection seed 42 was then used to select 100 correctly classified test nodes without replacement. This fixed set of target nodes was subsequently used to construct the explanation cohort and was not modified in response to later explanation or perturbation outcomes.

Baseline GraphLIME explanations were generated for all 100 selected target nodes. Valid explanations were obtained for 97 nodes. For the remaining three nodes, the local model-output distributions within their respective explanation neighborhoods were degenerate, preventing construction of an informative output kernel for GraphLIME. These nodes were retained in the experimental diagnostics rather than replaced by alternative correctly classified nodes. Consequently, the primary cohort for analyses requiring a valid baseline explanation consisted of 97 target nodes.

This selection procedure separates predictive correctness from explanation availability: nodes were selected according to the frozen GNN's test-set predictions before GraphLIME explainability was assessed. The resulting 97-node explanation cohort was therefore fixed prior to the controlled perturbation experiments. The graph structure, dataset splits, normalized baseline feature matrix, selected target-node set, and baseline explanation cohort were held fixed

throughout the subsequent stability analysis.

Graph Convolutional Network

A two-layer graph convolutional network (GCN) [1] was used as the predictive model whose local explanations were subsequently evaluated. The model consisted of two graph convolutional layers. The first layer transformed the input node features into a 16-dimensional hidden representation, followed by a rectified linear unit (ReLU) activation and dropout with probability 0.5. The second graph convolutional layer mapped the hidden representations to class logits. Model probabilities required by the explanation procedure were obtained by applying the softmax function to these logits outside the GCN itself.

Let $H^{(0)} = X$ denote the input node-feature matrix. Following the normalized graph-convolution formulation of Kipf and Welling [1], a graph convolutional layer can be expressed as

$$H^{(l+1)} = \sigma \left(\widetilde{D}^{-\frac{1}{2}} \widetilde{A} \widetilde{D}^{-\frac{1}{2}} H^{(l)} W^{(l)} \right), \quad (6)$$

where $\widetilde{A} = A + I$ denotes the adjacency matrix augmented with self-connections, $\widetilde{D}$ is the corresponding degree matrix, $W^{(l)}$ is the trainable weight matrix at layer l , and $\sigma(\cdot)$ denotes the layer's non-linear activation where applicable. The two-layer architecture allows information to propagate across the local graph neighborhood while producing node-level class scores for the citation-network classification task.

The GCN was trained for a maximum of 200 epochs using the Adam optimizer [14] with a learning rate of 0.01 and weight decay of 5×10^{-4} . Cross-entropy loss was evaluated on the predefined training nodes. Model selection was based exclusively on validation accuracy: after each training epoch, performance was evaluated on the predefined validation split, and the parameters associated with the highest observed validation accuracy were retained. In the event of an equal validation accuracy, the earlier checkpoint was preserved. The test split was not used for model selection.

Training and model selection were conducted with random seed 42. The selected checkpoint occurred at epoch 125, at which the model achieved a validation accuracy of 0.7840 and a training accuracy of 0.9929. After restoration of the selected parameters, the model achieved a test accuracy of 0.8060. This test performance is reported descriptively and did not influence checkpoint selection.

Following model selection, the GCN parameters were frozen for all subsequent explanation and perturbation experiments. The model was neither retrained nor fine-tuned after perturbations were introduced. Accordingly, changes observed in model predictions or GraphLIME explanations across experimental conditions arose from controlled changes to the input features rather than changes to the learned GCN parameters. The same frozen checkpoint was used to generate the baseline predictions, select the target nodes described in Section 3.2, construct GraphLIME explanations, and evaluate all perturbed observations.

GraphLIME Explanations

Local feature explanations were generated using GraphLIME [7], a post-hoc explanation framework for graph neural networks that constructs an interpretable model within the local graph neighbourhood of a target node. For each target node v , the local explanation domain was defined by its 2-hop graph neighbourhood. The corresponding node-feature vectors and predictions of the frozen GCN were extracted for the nodes in this neighbourhood. The GCN outputs used by GraphLIME were the complete class-probability vectors obtained by applying softmax to the model logits, rather than only the predicted class labels or target-class probabilities.

For a local neighbourhood containing n_v nodes, let $x^{(k)} \in \mathbb{R}^{n_v}$ denote the values of feature k across the neighbourhood and let $P_v \in \mathbb{R}^{n_v \times C}$ denote the corresponding GCN class-probability vectors for C classes. GraphLIME represents the dependence between each input feature and the model output through kernel matrices. A Gaussian kernel was constructed independently for every locally varying input feature. For neighbourhood nodes i and j , the kernel associated with feature k was

$$K_{ij}^{(k)} = \exp \left(-\frac{\left(x_i^{(k)} - x_j^{(k)} \right)^2}{2\sigma_k^2} \right), \quad (7)$$

where σ_k denotes the bandwidth associated with feature k . An analogous Gaussian kernel L was constructed from pairwise Euclidean distances between the full GCN probability vectors. In the implementation used in this study, each bandwidth was set to the median of the positive pairwise Euclidean distances for the corresponding local variable. Features that were constant within a target node's local neighbourhood were treated as locally degenerate and excluded from the active feature-kernel set. These bandwidth and degeneracy-handling rules were fixed implementation decisions used consistently throughout the experiments.

Each kernel was subsequently centred in the local sample space. Let

$$H = I_{n_v} - \frac{1}{n_v} \mathbf{1}\mathbf{1}^\top \quad (8)$$

denote the centring matrix. The centred feature and output kernels were obtained as $HK^{(k)}H$ and HLH , respectively. The implementation additionally normalized each non-degenerate centred kernel by its Frobenius norm. Denoting the resulting normalized kernels by $\bar{K}^{(k)}$ and $\bar{L}$, respectively, gives

$$\bar{K}^{(k)} = \frac{HK^{(k)}H}{\|HK^{(k)}H\|_F}, \quad \bar{L} = \frac{HLH}{\|HLH\|_F}. \quad (9)$$

The kernel-construction procedure, including centring, Frobenius normalization, finite-value checks, and exclusion of locally degenerate features, was validated before baseline explanations were generated.

Feature relevance was then estimated through the non-negative HSIC-Lasso [11] objective used by GraphLIME [7]. Given the normalized feature kernels and output kernel, the explanation coefficients β were obtained by solving

$$\min_{\beta \geq 0} \frac{1}{2} \|\bar{L} - \sum_{k=1}^{d_v} \beta_k \bar{K}^{(k)}\|_F^2 + \rho \|\beta\|_1, \quad (10)$$

where d_v is the number of locally non-degenerate features, β_k is the non-negative relevance coefficient associated with feature k , and ρ controls the sparsity of the explanation. Larger positive coefficients indicate features whose local kernel structure provides a stronger contribution to approximating the kernel representation of the GCN outputs.

The optimization problem in Eq. (10) was implemented by vectorizing the Frobenius-norm objective into its equivalent least-squares form and solving it using **non-negative proximal-gradient optimization**. Coefficients were initialized to zero, and each update combined a gradient step for the least-squares term with non-negative soft-thresholding for the l_1 penalty. The maximum number of optimization iterations was 10,000, with a relative coefficient-change convergence tolerance of 10^{-8} . Coefficients not exceeding 10^{-8} were treated as zero. This implementation solves the stated convex HSIC-Lasso objective but does **not** exactly reproduce the non-negative least-angle regression procedure used to optimize the objective in the original GraphLIME implementation [7]; the distinction is retained explicitly when interpreting the present study as an objective-

faithful implementation rather than an exact optimizer-level reproduction.

The regularization parameter was fixed at $\rho = 0.03$ before the perturbation experiments. It was selected using a predetermined calibration subset consisting of the first 10 nodes from the frozen target-node selection and candidate values $\{0.001, 0.003, 0.01, 0.03, 0.1, 0.3\}$. The calibration procedure required optimizer convergence and sufficient non-zero coefficients to construct a complete top-10 explanation for every calibration node. The remaining selected nodes and all subsequent perturbation results were not used to retune ρ .

For each target node admitting a valid output kernel and converged solution, features with positive coefficients above the fixed zero threshold were ranked by decreasing β_k , and the 10 highest-ranked features formed the explanation set $E_0^{(10)}(v)$ defined in Section 3.1. The frozen configuration therefore used a 2-hop neighbourhood, $\rho = 0.03$, and $K = 10$ throughout the baseline and perturbation experiments. Under this configuration, valid baseline output kernels were obtained for 97 of the 100 predetermined target nodes, and all 97 corresponding optimization problems converged with at least 10 non-zero coefficients, yielding complete top-10 baseline explanations. The three output-degenerate cases were retained as diagnostic outcomes rather than replaced.

Controlled Feature Perturbation

To evaluate the stability of GraphLIME explanations under controlled changes to the input, feature perturbations were applied to the local neighbourhood of each target node while all other components of the predictive system were held fixed. The perturbation procedure modified only the node-feature matrix: the graph topology and the parameters of the trained GCN remained unchanged, and the model was not retrained or fine-tuned after perturbation. This design isolates changes in predictions and explanations that arise from controlled modification of the input features rather than from changes to graph structure or learned model parameters.

For a target node v , perturbations were restricted to the same 2-hop neighbourhood used to construct its GraphLIME explanation. Let $\mathcal{A}_v$ denote the set of active local feature entries,

$$\mathcal{A}_v = \{(i, k), : i \in \mathcal{N}_2(v) \text{ } X_{ik} \neq 0\}, \quad (11)$$

where $\mathcal{N}_2(v)$ denotes the set of nodes in the extracted 2-hop neighbourhood of v , including the target node, and X_{ik} is the value of feature k for node i . Only entries that were non-zero in the original

feature matrix were eligible for masking. Consequently, the perturbation operator removed locally active feature information rather than selecting entries that were already zero.

Four masking rates were evaluated,

$$\mathcal{R} = \{0.01, 0.05, 0.10, 0.20\}, \quad (12)$$

corresponding to nominal masking levels of 1%, 5%, 10%, and 20% of the active entries in the local neighbourhood. For a target node with $n_v^{active} = |\mathcal{A}_v|$ eligible entries, the number of entries masked at rate r was determined by

$$m_v(r) = \min \{n_v^{active}, \max(1, \lfloor r n_v^{active} + 0.5 \rfloor)\}. \quad (13)$$

The lower bound of one ensures that every nominal perturbation condition produces an actual feature modification, while the upper bound prevents the requested mask size from exceeding the number of eligible entries.

For each target node, masking rate, and perturbation seed, $m_v(r)$ entries were sampled without replacement from $\mathcal{A}_v$ and their values were set to zero. Let $\mathcal{M}_{v,r,s} \subseteq \mathcal{A}_v$ denote the sampled mask for target node v , rate r , and perturbation seed s . The perturbed feature matrix can then be written element-wise as

$$[X_{r,s}^{(v)}]_{ik} = \begin{cases} 0, & (i, k) \in \mathcal{M}_{v,r,s}, \\ X_{ik}, & \text{otherwise.} \end{cases} \quad (14)$$

Each perturbation observation was constructed from a fresh copy of the original normalized feature matrix X . Perturbations were therefore not accumulated across masking rates or repetitions. No feature renormalization was performed after masking. Masks for different rates were sampled separately rather than constructed as nested subsets; consequently, the 5% condition, for example, was not required to contain the entries masked in the 1% condition.

Ten predetermined perturbation seeds, $s \in \{101, \dots, 110\}$, were evaluated at each masking rate. Random mask generation was deterministic for a given target node, masking rate, and perturbation seed through a derived random seed incorporating all three quantities. This provided reproducible perturbation realizations while allowing repeated masks to be evaluated within each node-rate condition.

The perturbation experiment was performed on the 97 target nodes with valid baseline GraphLIME explanations identified in Section

3.2. Combining 97 nodes, four masking rates, and ten perturbation repetitions produced

$$97 \times 4 \times 10 = 3880 \quad (15)$$

perturbed observations. For each observation, the frozen GCN was evaluated on the perturbed feature matrix, after which GraphLIME was recomputed using the perturbed input and the unchanged graph. The resulting prediction, model confidence, explanation coefficients, top-10 feature set when available, and explanation-status diagnostics were recorded for subsequent stability analysis.

This protocol therefore varied a single experimental factor – the extent of local active-feature masking – while preserving the graph structure, trained model parameters, target-node cohort, explanation configuration, and all other frozen experimental settings. The resulting observations provide the controlled perturbation basis for the stability metrics defined in Section 3.6.

Stability Metrics

The effect of controlled feature masking was evaluated using complementary measures of predictive and explanatory stability. For each target node v , masking rate r , and perturbation seed s , the perturbed observation was compared with the corresponding unperturbed baseline. The analysis considered four quantities: prediction agreement, confidence change, top-10 explanation overlap, and explanation availability. These measures were evaluated separately because preservation of the model output does not necessarily imply preservation of the explanation.

Prediction agreement

Prediction stability was measured by whether the predicted class remained unchanged following perturbation. As introduced in Eq. (3), observation-level prediction agreement was defined as

$$P_{r,s}(v) = \mathbb{I} [\hat{y}_{r,s}(v) = \hat{y}_0(v)] , \quad (16)$$

where $P_{r,s}(v) = 1$ indicates that the perturbed and baseline predictions agree and $P_{r,s}(v) = 0$ indicates a change in predicted class. For a fixed target node and masking rate, prediction stability was summarized across the ten perturbation repetitions as

$$\overline{P}_r(v) = \frac{1}{S} \sum_{s=1}^S P_{r,s}(v), \quad S = 10. \quad (17)$$

Thus, $\bar{P}_r(v) \in [0, 1]$ represents the proportion of perturbation repetitions at rate r for which the target node retained its baseline predicted class.

Confidence change

Changes in predictive confidence were recorded separately from class agreement. Let

$$C_0(v) = \max_c [p_0(v)]_c$$

denote the maximum class probability assigned by the baseline model, and let

$$C_{r,s}(v) = \max_c [p_{r,s}(v)]_c$$

denote the corresponding maximum probability after perturbation. Confidence change was defined as

$$\Delta C_{r,s}(v) = C_{r,s}(v) - C_0(v). \quad (18)$$

A negative value therefore indicates a reduction in the model's maximum predicted probability relative to baseline, whereas a positive value indicates an increase. Confidence change was retained as a complementary diagnostic because the predicted class may remain unchanged even when the model's probability distribution changes.

Explanation overlap

Explanation stability was quantified by comparing the top-10 GraphLIME feature set obtained under perturbation with the corresponding baseline explanation. For observations in which a valid perturbed explanation was available, the top- K Jaccard similarity [15] introduced in Eq. (4) was evaluated with $K = 10$:

$$J_{r,s}^{(10)}(v) = \frac{|E_0^{(10)}(v) \cap E_{r,s}^{(10)}(v)|}{|E_0^{(10)}(v) \cup E_{r,s}^{(10)}(v)|}. \quad (19)$$

The measure lies in $[0, 1]$, where larger values indicate greater overlap between the baseline and perturbed top-10 feature sets. A value of one indicates identical sets, whereas a value of zero indicates that the two valid top-10 explanations share no selected features.

For a target node and masking rate, explanation stability was summarized across perturbation repetitions for which a valid perturbed explanation existed. Let

$$\mathcal{S}_r^{valid}(v) = \left\{ s : E_{r,s}^{(10)}(v) \text{ is available} \right\}.$$

The node-level mean explanation stability was then

$$\bar{J}_r(v) = \frac{1}{|\mathcal{S}_r^{valid}(v)|} \sum_{s \in \mathcal{S}_r^{valid}(v)} J_{r,s}^{(10)}(v), \quad (20)$$

provided that at least one valid perturbed explanation was available for the corresponding node-rate condition.

Because the principal research question concerns whether explanations can change even when the model retains its decision, explanation overlap was additionally evaluated **conditional on prediction agreement**. Define

$$\mathcal{S}_r^{same}(v) = \left\{ s : P_{r,s}(v) = 1 \wedge E_{r,s}^{(10)}(v) \text{ is available} \right\}.$$

Conditional explanation stability was summarized as

$$\bar{J}_r^{same}(v) = \frac{1}{|\mathcal{S}_r^{same}(v)|} \sum_{s \in \mathcal{S}_r^{same}(v)} J_{r,s}^{(10)}(v), \quad (21)$$

when the conditioning set was non-empty. This quantity measures explanation overlap specifically among perturbation repetitions for which the GCN's predicted class remained unchanged.

Explanation availability

Explanation availability was treated as a distinct outcome rather than incorporated into the Jaccard score. For each perturbed observation, define

$$A_{r,s}(v) = \mathbb{I} \left[E_{r,s}^{(10)}(v) \text{ is available} \right]. \quad (22)$$

A perturbed explanation was considered available when the GraphLIME procedure produced a valid solution from which a complete top-10 feature set could be constructed. If this condition was not satisfied, the explanation was recorded as unavailable and the corresponding Jaccard similarity was treated as missing.

In particular, an unavailable explanation was **not assigned a Jaccard similarity of zero**. A zero Jaccard score has a distinct interpretation: it represents two valid explanation sets with no overlapping selected features. Conflating explanation failure with zero overlap would therefore combine two qualitatively different outcomes and systematically distort the explanation-stability measure. Explanation availability was instead summarized independently as

$$\bar{A}_r(v) = \frac{1}{S} \sum_{s=1}^S A_{r,s}(v), \quad S = 10. \quad (23)$$

Together, these metrics distinguish whether a perturbation changes the predicted class, changes predictive confidence, alters the selected explanatory features, or prevents a complete GraphLIME explanation from being obtained. This separation is central to the analysis because predictive robustness, explanation stability, and explanation availability represent related but non-equivalent properties of the explanatory system. The statistical procedures used to aggregate and compare these quantities across target nodes and masking rates are described in Section 3.7.

Statistical Analysis

The statistical analysis was designed to preserve the repeated-measures structure of the perturbation experiment. Although ten perturbation realizations were generated for each target node at each masking rate, these realizations were treated as repeated measurements within a node rather than as independent statistical observations. The **target node** was therefore taken as the primary statistical unit. Observation-level measurements were first aggregated within each node-rate condition before comparisons across masking rates were performed.

For each masking rate r , the node-level quantities defined in Section 3.6 were summarized across the 97 target nodes in the explanation cohort. Let $M_r(v)$ denote a generic node-level stability measure for target node v , such as mean prediction agreement $\bar{P}_r(v)$ or mean explanation overlap $\bar{J}_r(v)$. The corresponding across-node mean was calculated as

$$\bar{M}_r = \frac{1}{N_r} \sum_{v \in \mathcal{V}_r} M_r(v), \quad (24)$$

where $\mathcal{V}_r$ is the set of target nodes for which the relevant node-level measure is defined at masking rate r , and $N_r = |\mathcal{V}_r|$. For measures requiring valid perturbed explanations, unavailable explanations remained missing as specified in Section 3.6 and were not converted to numerical stability scores.

Uncertainty around across-node means was quantified using **95% node-level bootstrap confidence intervals** [16]. Target nodes, rather than individual perturbation observations, were resampled with replacement so that the repeated perturbation measurements belonging to a node remained grouped within the statistical unit. A total of 10,000 bootstrap resamples were generated using bootstrap

seed 20260917. Percentile confidence intervals were obtained from the 2.5th and 97.5th percentiles of the resulting bootstrap distribution.

Changes across masking rates were evaluated using paired node-level contrasts. For two masking rates r_a and r_b , the within-node difference for a stability measure M was defined as

$$D_{r_a, r_b}(v) = M_{r_b}(v) - M_{r_a}(v), \quad (25)$$

for nodes with the relevant measure available at both rates. The mean paired difference was

$$\bar{D}_{r_a, r_b} = \frac{1}{N_{ab}} \sum_{v \in \mathcal{V}_{ab}} D_{r_a, r_b}(v), \quad (26)$$

where $\mathcal{V}_{ab}$ denotes the paired set of eligible target nodes and $N_{ab} = |\mathcal{V}_{ab}|$. Confidence intervals for paired differences were obtained by bootstrapping the target nodes and retaining each node's paired measurements together. The principal endpoint contrasts compared the lowest masking rate, $r = 0.01$, with the highest masking rate, $r = 0.20$, while the complete analysis also retained the intermediate masking conditions.

As a secondary inferential analysis, paired differences were evaluated using two-sided **Wilcoxon signed-rank tests** [17]. The analysis used a NumPy/Python implementation with a tie-corrected normal approximation and without continuity correction. Zero paired differences were excluded from the signed-rank calculation. For each outcome family containing the six pairwise contrasts among the four masking rates, multiplicity was controlled using the Holm procedure [18]. Effect magnitude and direction were additionally summarized using the matched-pairs rank-biserial correlation [19],

$$r_{rb} = \frac{W^+ - W^-}{W^+ + W^-}, \quad (27)$$

where W^+ and W^- denote the sums of ranks associated with positive and negative non-zero paired differences, respectively. Under the difference convention in Eq. (25), the sign of r_{rb} indicates the direction of change from the lower-rate condition to the higher-rate condition.

In addition to pairwise contrasts, the consistency of rate-dependent changes was assessed at the target-node level. For explanation stability, prediction-conditioned explanation stability, and prediction stability, the sequence of node-level means across the ordered masking

rates 0.01, 0.05, 0.10, and 0.20 was examined. The proportion of eligible target nodes exhibiting a monotonically non-increasing sequence was recorded as a descriptive measure of whether the aggregate direction of change was also reflected consistently across individual nodes.

The relationship between prediction and explanation stability was analysed separately to address whether the two quantities were coupled across target nodes. For each node with valid measurements at both endpoint masking rates, endpoint changes were calculated as

$$\Delta P(v) = \bar{P}_{0.20}(v) - \bar{P}_{0.01}(v), \quad \Delta J(v) = \bar{J}_{0.20}(v) - \bar{J}_{0.01}(v). \quad (28)$$

Pearson and Spearman correlation coefficients between $\Delta P(v)$ and $\Delta J(v)$ were calculated as **descriptive measures of coupling**. No independent-and-identically-distributed correlation significance test was used for this analysis. The analysis additionally recorded the number and proportion of target nodes whose endpoint prediction-stability proportion was unchanged while their explanation overlap decreased. Here, an unchanged endpoint prediction-stability proportion means that the proportion of the ten perturbation repetitions preserving the predicted class was equal at the 1% and 20% masking conditions; it does not imply that every individual perturbation realization produced an unchanged prediction.

Explanation availability was analysed independently of explanation overlap. Availability counts and proportions were calculated at each masking rate from the ten perturbation repetitions per target node. The rate-wise pattern of unavailable explanations was treated descriptively rather than interpreted as a numerical explanation-stability score. This maintained the distinction between failure to obtain a complete GraphLIME explanation and obtaining two valid explanations whose selected feature sets differ.

The analysis was organized around five hypotheses specified before final statistical interpretation. H1 examined whether top-10 explanation overlap decreased as masking severity increased. H2 examined the same relationship conditional on preservation of the predicted class. H3 examined whether prediction agreement decreased with increasing masking severity. H4 examined whether changes in prediction stability and explanation stability were imperfectly coupled across target nodes. H5 examined whether explanation unavailability increased with perturbation severity. The statistical procedures described above were applied consistently to these hypotheses without post-hoc modification of the frozen perturbation protocol.

Finally, the inferential scope of the analysis is constrained by the graph structure of the experiment. All target nodes originate from a **single Cora graph**, and graph neighbourhoods may overlap; consequently, target-node measurements cannot be assumed to represent independent samples drawn from a broader population of graphs. Node-level bootstrap confidence intervals and paired tests preserve the experimental unit more appropriately than treating the 3,880 perturbation observations as independent, but they do not eliminate dependence induced by the shared graph. The resulting statistical quantities are therefore interpreted as uncertainty and paired evidence within this fixed experimental graph rather than as unrestricted population-level inference across arbitrary graphs or datasets. This limitation is considered explicitly in Section 6.

Results

Baseline Model and Explanation Validation

The selected GCN checkpoint was obtained at epoch 125, with a training accuracy of 0.9929 and a validation accuracy of 0.7840. After restoration of this checkpoint, the model achieved a test accuracy of 0.8060. As specified in Section 3.3, test performance was evaluated only after validation-based model selection and did not contribute to checkpoint selection.

From the correctly classified test nodes, 100 target nodes were selected using the predetermined selection procedure described in Section 3.2. Baseline GraphLIME processing produced valid output kernels for 97 of these nodes. For the remaining three target nodes (node IDs 2697, 2625, and 2411), the GCN probability vectors within the corresponding local explanation neighbourhoods exhibited insufficient variation to construct a non-degenerate output kernel. These cases were therefore classified as output-degenerate. They were retained in the diagnostic record and were not replaced by alternative test nodes. Consequently, the subsequent explanation-stability experiments were conducted on the fixed cohort of 97 baseline-explainable target nodes.

For all 97 nodes in this cohort, the GraphLIME optimization procedure converged and produced at least 10 non-zero feature coefficients, permitting a complete top-10 explanation to be constructed for every baseline-explainable node. The numerical solutions were additionally evaluated using the Karush–Kuhn–Tucker (KKT) optimality conditions for the constrained HSIC-Lasso objective. All 97 solutions passed the predefined KKT validation criterion. Across the validated solutions, the maximum active-set stationarity resid-

ual was 6.59×10^{-7} , the maximum inactive-set violation was zero, the minimum inactive-set slack was 2.80×10^{-5} , and the maximum complementarity residual was 3.79×10^{-7} . These quantities were below the corresponding validation tolerance where applicable and provided numerical confirmation that the retained baseline coefficient solutions satisfied the required first-order optimality conditions.

Thus, the baseline stage yielded a frozen GCN checkpoint, a predetermined set of 100 correctly classified target nodes, and 97 target nodes for which complete and numerically validated baseline GraphLIME explanations were available. This 97-node cohort formed the fixed reference population for the controlled feature-perturbation analyses reported in Sections 4.2–4.5.

Table 1. Baseline model and GraphLIME validation summary

Quantity	Result
Selected checkpoint epoch	125
Training accuracy	0.9929
Validation accuracy	0.7840
Test accuracy	0.8060
Predetermined target nodes	100
Valid baseline GraphLIME explanations	97
Output-degenerate cases	3
Complete top-10 explanations	97/97
KKT validation passes	97/97
Maximum active-set stationarity residual	6.59×10^{-7}
Maximum inactive-set violation	0
Minimum inactive-set slack	2.80×10^{-5}
Maximum complementarity residual	3.79×10^{-7}

Table 1 Baseline predictive performance, explanation availability, and numerical validation of the GraphLIME solutions. Model selection was based on validation accuracy; test accuracy was evaluated only after checkpoint selection. KKT diagnostics are reported across the 97 valid baseline GraphLIME solutions.

Explanation Stability under Feature Masking

GraphLIME explanation stability decreased progressively as the severity of feature masking increased. Across target nodes with valid perturbed explanations, the mean top-10 Jaccard similarity between baseline and perturbed explanations was **0.9015** at the 1% masking rate, **0.7253** at 5%, **0.6045** at 10%, and **0.4541** at 20%.

Thus, even though explanations remained highly similar to baseline under the smallest perturbation, progressively larger fractions of the selected top-10 features changed as the masking rate increased (Fig. 1).

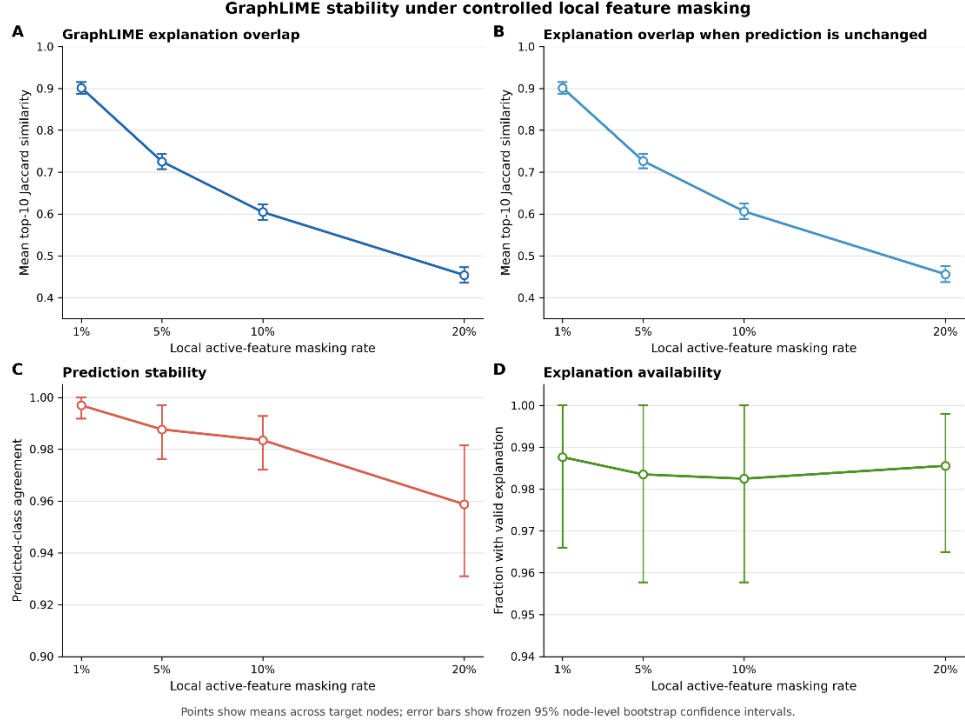


Fig. 1 Explanation and prediction stability across feature-masking rates. Points show means across target nodes; error bars show frozen 95% node-level bootstrap confidence intervals.

The reduction in explanation overlap was apparent across the complete range of masking conditions rather than being confined to the most severe perturbation. Relative to the mean Jaccard similarity of 0.9015 at 1% masking, the mean had decreased to 0.7253 at 5% and 0.6045 at 10%, before reaching 0.4541 at 20%. At the highest masking rate, the mean top-10 explanation therefore shared substantially less of its selected feature set with the corresponding baseline explanation than under the lowest masking condition. Figure 1 summarizes this rate-dependent pattern together with the frozen 95% node-level bootstrap confidence intervals.

The paired endpoint analysis provided further evidence for this reduction. Comparing the 1% and 20% masking conditions within target nodes, the mean paired change in top-10 Jaccard similarity

was

$$\Delta J_{0.01,0.20} = -0.4475, \quad (29)$$

with a 95% node-level bootstrap confidence interval of **[-0.4660, -0.4282]**. The corresponding two-sided Wilcoxon signed-rank comparison remained significant after Holm adjustment ($p_{Holm} = 7.30 \times 10^{-17}$), with a matched-pairs rank-biserial correlation of **-1.00** under the difference convention defined in Section 3.7. The negative direction indicates lower explanation overlap at 20% masking than at 1% masking among the non-zero paired differences included in the signed-rank calculation.

The rate-dependent pattern was also evident at the level of individual target nodes. Across the ordered masking rates of 1%, 5%, 10%, and 20%, **93.75%** of eligible nodes exhibited a monotonically non-increasing sequence of mean Jaccard similarity. The aggregate decline in explanation overlap was therefore accompanied by a highly consistent directional pattern across the target-node cohort, although strict monotonicity was not observed for every node.

Taken together, these results support **H1**: top-10 GraphLIME explanation overlap decreased as the severity of controlled feature masking increased. The effect was substantial between the endpoint conditions and was observed consistently across most target nodes. Whether this decline persists when analysis is restricted specifically to perturbations that preserve the GCN’s predicted class is examined in Section 4.3.

Stability Conditional on Prediction Agreement

The reduction in GraphLIME explanation stability persisted when the analysis was restricted to perturbation observations for which the GCN retained its baseline predicted class. Among observations satisfying both prediction agreement and explanation availability, the mean top-10 Jaccard similarity was **0.9014** at 1% masking, **0.7266** at 5%, **0.6063** at 10%, and **0.4560** at 20%. The prediction-conditioned trajectory therefore closely followed the decline observed in the unconditional explanation-stability analysis of Section 4.2.

The endpoint comparison showed a substantial decrease in explanation overlap despite preservation of the predicted class. Between the 1% and 20% masking conditions, the mean paired change in prediction-conditioned Jaccard similarity was

$$\Delta J_{0.01,0.20}^{same} = -0.4454, \quad (30)$$

with a 95% node-level bootstrap confidence interval of **[-0.4637, -0.4269]**. The corresponding two-sided Wilcoxon signed-rank comparison remained significant after Holm adjustment ($p_{Holm} = 7.30 \times 10^{-17}$), with a matched-pairs rank-biserial correlation of **-1.00** under the difference convention specified in Section 3.7.

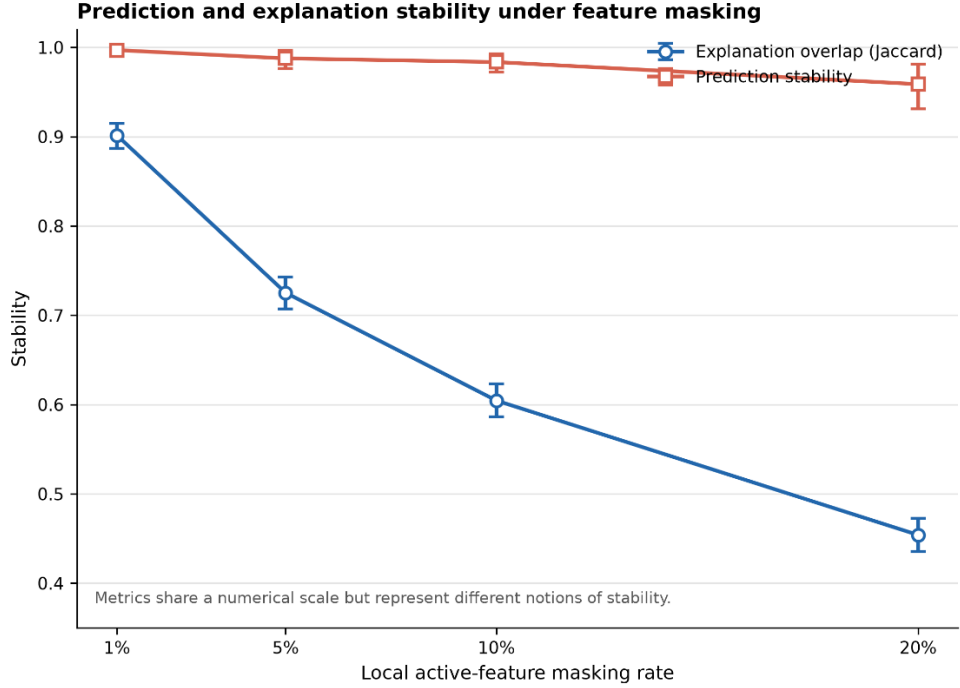


Fig. 2 Prediction and explanation stability under controlled feature masking. Explanation overlap decreases substantially with increasing masking severity despite comparatively high prediction stability, illustrating that preservation of the predicted class does not imply preservation of the corresponding GraphLIME explanation.

The directional pattern was also consistent across individual target nodes. Across the ordered masking conditions, **92.71%** of eligible nodes exhibited a monotonically non-increasing sequence of mean prediction-conditioned Jaccard similarity. As in the unconditional analysis, this proportion indicates a highly consistent decline across the cohort while allowing for node-specific departures from strict monotonicity.

These results support **H2** and demonstrate that the decline in explanation overlap cannot be accounted for solely by perturbations that change the model's predicted class. Even when analysis was restricted to observations satisfying

$$\hat{y}_{r,s}(v) = \hat{y}_0(v), \quad (31)$$

the features selected by GraphLIME became progressively less similar to the baseline explanation as masking severity increased. Preservation of the categorical prediction was therefore compatible with substantial changes in the corresponding local feature explanation.

The distinction is also visible when the prediction-conditioned and unconditional explanation-stability trajectories are compared. At 20% masking, for example, the unconditional mean Jaccard similarity was **0.4541**, whereas the prediction-conditioned mean was **0.4560**. The small numerical difference between these summaries indicates that restricting the analysis to prediction-preserving observations did not materially alter the observed rate-dependent decline in explanation overlap. This comparison is descriptive; it was not treated as a separate inferential hypothesis.

The results therefore establish the central empirical phenomenon motivating the study: **prediction preservation does not imply explanation preservation** under the controlled feature perturbations considered here. The extent to which changes in prediction stability and explanation stability are coupled across target nodes is examined directly in Section 4.4.

Prediction–Explanation Stability Coupling

The preceding analyses showed that explanation overlap decreased substantially with increasing feature masking, including when the GCN retained its baseline predicted class. To examine whether changes in predictive and explanatory stability were coupled at the target-node level, endpoint changes between the 1% and 20% masking conditions were compared for the 97 nodes with valid measurements at both endpoints.

Across these nodes, mean prediction stability decreased by **0.0381** between the endpoint conditions, whereas mean explanation stability decreased by **0.4475**. Thus, the magnitude of the observed change in top-10 explanation overlap was considerably larger than the corresponding change in the proportion of perturbation repetitions preserving the predicted class. These quantities measure different outcomes and are not interpreted as directly commensurate effect sizes; rather, their contrasting endpoint behaviour motivates examination of their node-level association.

The association between the two endpoint changes was weak. As defined in Section 3.7, let $\Delta P(v)$ denote the change in prediction stability and $\Delta J(v)$ the change in explanation stability between 1% and

20% masking. Across the 97 endpoint-valid nodes, the descriptive Spearman correlation was

$$\rho_S(\Delta, P\Delta J) = -0.0271, \quad (32)$$

while the descriptive Pearson correlation was

$$r(\Delta, P\Delta J) = 0.0927. \quad (33)$$

Both coefficients were close to zero, indicating little observed node-level correspondence between the magnitude of change in prediction stability and the magnitude of change in explanation overlap within this experiment. As specified in Section 3.7, these correlations were treated descriptively; no independent-and-identically-distributed correlation significance test was applied because the target nodes originate from a single graph and their neighbourhoods may overlap.

A particularly prominent form of this divergence occurred among nodes whose endpoint prediction-stability proportion did not change. **Eighty-six of the 97 nodes (88.66%)** had the same prediction-stability proportion at 1% and 20% masking while simultaneously exhibiting a decrease in mean explanation overlap. In this analysis, an unchanged endpoint prediction-stability proportion means that the fraction of the ten perturbation repetitions preserving the baseline predicted class was equal at the two endpoint masking rates. It does **not** imply that every individual perturbation realization preserved the prediction or that identical perturbation outcomes occurred at both rates.

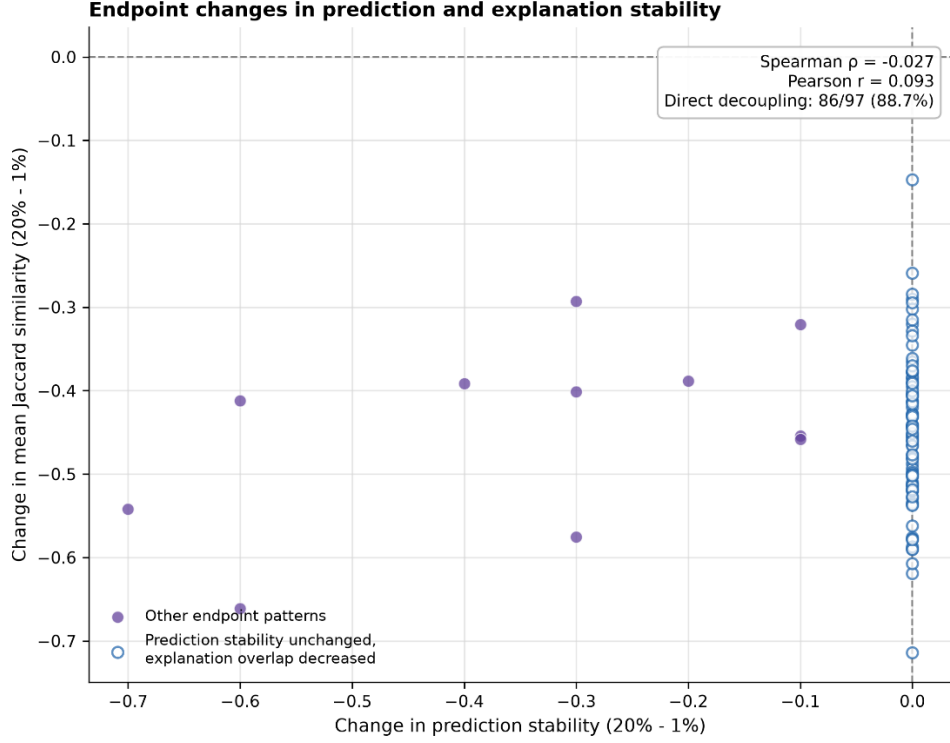


Fig. 3 Endpoint coupling between prediction and explanation stability from 1% to 20% feature masking. Each point represents a target node and shows the change in prediction-stability proportion against the corresponding change in mean top-10 Jaccard explanation overlap. The vertical concentration at zero indicates nodes whose endpoint prediction-stability proportion was unchanged, while negative explanation-stability changes indicate reduced explanation overlap.

Figure 3 visualizes the joint endpoint changes in prediction and explanation stability. The concentration of observations around $\Delta P(v) = 0$, together with predominantly negative values of $\Delta J(v)$, illustrates the principal empirical pattern: for many target nodes, explanation overlap declined even though the aggregate proportion of prediction-preserving perturbations remained unchanged between the endpoint conditions.

These findings support **H4**, which proposed that prediction and explanation stability would not be perfectly coupled. Within the fixed Cora experiment, changes in prediction stability provided little indication of the corresponding magnitude of changes in GraphLIME explanation overlap. The result complements the prediction-conditioned analysis in Section 4.3 by showing at the

target-node level that explanatory stability can deteriorate substantially without a corresponding deterioration in the measured proportion of stable predictions.

Explanation Availability

GraphLIME explanation availability remained high across all four feature-masking conditions. Of the **3,880** perturbation observations, **3,821 (98.48%)** produced valid perturbed explanations, while **59 (1.52%)** were unavailable. As specified in Section 3.6, unavailable explanations were analysed separately and their Jaccard similarities were treated as missing rather than assigned a value of zero.

The number of unavailable explanations was **12** at 1% masking, **16** at 5%, **17** at 10%, and **14** at 20%. Because each masking condition contained 970 perturbation observations, these counts correspond to unavailability rates of approximately **1.24%**, **1.65%**, **1.75%**, and **1.44%**, respectively. Equivalently, explanation availability was approximately **98.76%**, **98.35%**, **98.25%**, and **98.56%** across the four masking rates. Figure 4 summarizes the rate-specific pattern.

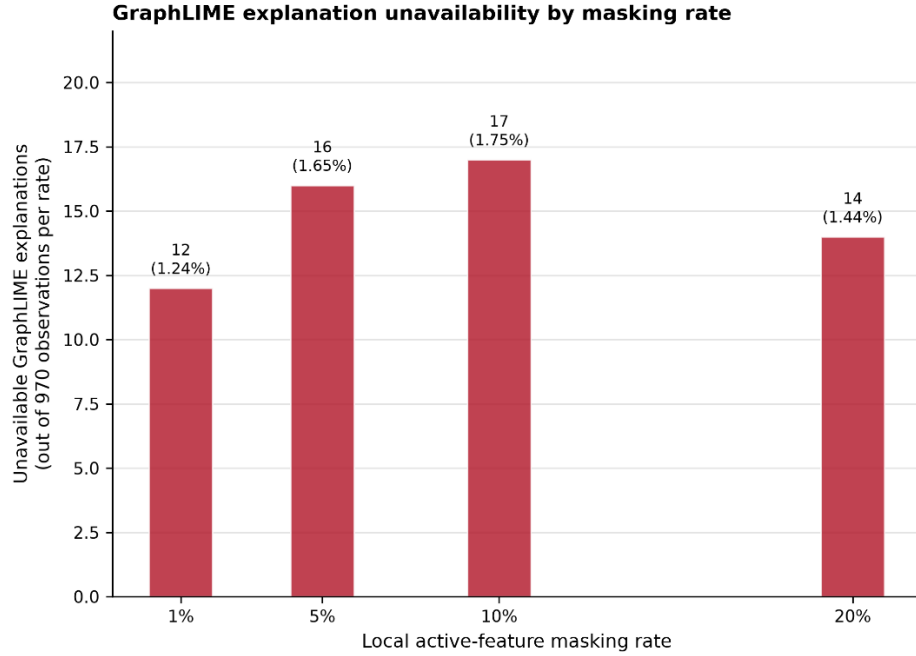


Fig. 4 GraphLIME explanation unavailability across feature-masking rates. Bars show the proportion of perturbation observations for which a complete top-10 perturbed explanation was unavailable. Unavailability remained low across all masking

conditions and did not increase monotonically with masking severity.

Importantly, explanation unavailability did **not** increase monotonically with masking severity. Although the number of unavailable explanations increased from 12 at 1% masking to 17 at 10%, it subsequently decreased to 14 at 20%. The observed rate-wise pattern therefore did not support the directional expectation specified in **H5** that explanation unavailability would increase as perturbation severity increased.

The 59 unavailable perturbed explanations were associated with two recorded failure modes. In **35 observations**, the fitted GraphLIME solution contained fewer than the 10 non-zero coefficients required to construct a complete top-10 explanation. The remaining **24 observations** were classified as optimizer non-convergence. These cases were retained in the diagnostic record and were not converted into artificial explanation-overlap values.

The distinction between explanation unavailability and explanation disagreement is important for interpreting the preceding stability results. A valid perturbed explanation with low Jaccard similarity indicates that GraphLIME successfully produced an explanation whose selected features differed substantially from the baseline set. An unavailable explanation instead indicates that a complete top-10 explanation could not be obtained under the predefined validity criteria. Treating both outcomes as $J = 0$ would therefore conflate explanation change with explanation failure. The high overall availability observed here indicates that the declining Jaccard similarities reported in Sections 4.2-4.4 were calculated predominantly from successfully obtained perturbed explanations rather than being driven by unavailable cases.

Overall, explanation availability remained above **98% at every masking rate**, while explanation overlap declined substantially as masking severity increased. The experiment therefore provided mixed evidence across the stability dimensions considered: feature masking produced a pronounced rate-dependent reduction in explanation overlap, but it did not produce a corresponding monotonic deterioration in the ability of GraphLIME to return a complete explanation. Accordingly, **H5 was not supported by the observed rate-wise availability pattern**.

Table 2. Stability outcomes across feature-masking rates

Masking rate	Prediction stability	Explanation availability	Mean top-10 Jaccard	Mean Jaccard given prediction agreement
1%	0.9969	0.9876	0.9015	0.9014
5%	0.9876	0.9835	0.7253	0.7266
10%	0.9835	0.9825	0.6045	0.6063
20%	0.9588	0.9856	0.4541	0.4560

Table 2 Summary of predictive and explanatory stability across feature-masking conditions. Prediction stability denotes the proportion of perturbation observations preserving the baseline predicted class. Explanation availability denotes the proportion yielding a valid complete top-10 GraphLIME explanation. Jaccard similarity is calculated only for observations with valid explanations; unavailable explanations are not assigned zero similarity.

Discussion

This study investigated whether local GraphLIME feature explanations remain stable when the input features of a frozen GCN are subjected to controlled perturbations. The principal finding is that predictive stability and explanation stability exhibited markedly different behaviour under increasing feature masking. Across the four perturbation levels, prediction agreement remained comparatively high, decreasing from 0.9969 at 1% masking to 0.9588 at 20%, whereas mean top-10 explanation overlap decreased from 0.9015 to 0.4541. The prediction-conditioned analysis produced a nearly identical decline in explanation overlap, from 0.9014 to 0.4560. These results indicate that, within the experimental conditions considered here, preservation of the model's categorical prediction did not imply preservation of the corresponding local feature explanation.

The observed decline in explanation overlap supports H1 and shows that GraphLIME explanations became progressively less stable as the magnitude of feature perturbation increased. The endpoint decrease of 0.4475 in mean top-10 Jaccard similarity was accompanied by a 95% bootstrap confidence interval that excluded zero, and 93.75% of eligible target nodes exhibited a monotonically non-increasing explanation-stability trajectory across the ordered masking rates. This pattern suggests that the aggregate decline was not attributable to a small number of unusually unstable nodes. Instead, decreasing overlap was observed across a large majority of the evaluated target-node cohort. The result is relevant to the broader eval-

uation of post-hoc GNN explanations because robustness to small changes in the explained input has been identified as an important property of reliable explanation methods [2, 3]. Recent surveys of GNN explainability similarly distinguish explanation robustness and stability from conventional measures of predictive performance [2].

The prediction-conditioned results provide an important qualification to this finding. Under H2, explanation overlap was evaluated only for perturbation observations in which the GCN retained its baseline predicted class. Nevertheless, mean Jaccard similarity declined from 0.9014 at 1% masking to 0.4560 at 20%, with a mean paired endpoint change of -0.4454. This decline was almost identical in magnitude to that obtained without conditioning on prediction agreement. Consequently, explanation instability in this experiment cannot be attributed solely to cases in which feature perturbation caused the model to cross a decision boundary and change its predicted class. Instead, substantial changes in the selected explanatory features were observed even among perturbations for which the categorical model decision was preserved.

This distinction is consistent with a broader concern in explanation robustness research: predictive invariance and explanatory invariance are different properties. Previous work has shown that explanations of neural models can be sensitive to small input perturbations, motivating robustness and stability as evaluation dimensions in their own right [2, 3]. In the graph domain, Li et al. demonstrated that GNN explanations can change substantially under limited structural perturbations while model predictions remain correct, illustrating that predictive behaviour alone does not guarantee explanation robustness [9]. The present study examines a different setting – non-adversarial feature masking rather than adversarial structural modification – but the observed separation between prediction preservation and explanation preservation is compatible with this broader distinction. The current results should therefore be interpreted as evidence for this phenomenon specifically within the controlled GraphLIME experiment, rather than as a direct replication of adversarial robustness results.

The prediction-stability results under H3 further clarify this separation. Prediction agreement decreased with masking severity, with a mean endpoint change of -0.0381 between 1% and 20% masking. Thus, the perturbations were not entirely behaviourally neutral: stronger feature masking produced some deterioration in prediction stability. However, the scale of the observed change differed substantially from the simultaneous decrease in explanation overlap. At 20% masking, prediction stability remained 0.9588, while mean explanation overlap had fallen to 0.4541. These values should not

be interpreted as directly comparable effect sizes because they quantify different properties. They nevertheless demonstrate that high aggregate prediction agreement can coexist with substantially lower agreement between the features selected by the local explanation procedure.

The node-level coupling analysis under H4 provides additional evidence that the two forms of stability were not tightly linked in this experiment. Across the 97 endpoint-valid target nodes, the descriptive Spearman correlation between endpoint changes in prediction and explanation stability was -0.0271, while the Pearson correlation was 0.0927. Both values were close to zero. Moreover, 86 of 97 nodes (88.66%) had an unchanged prediction-stability proportion between the 1% and 20% masking conditions while exhibiting decreased explanation overlap. The latter result is particularly informative because it demonstrates that a node need not show a deterioration in its aggregate rate of prediction preservation for its explanation overlap to decline. As emphasized in Section 4.4, however, equality of endpoint prediction-stability proportions does not imply identical prediction outcomes for every perturbation realization.

Taken together, H1-H4 therefore point to a common empirical conclusion: explanation stability constituted a distinct behaviour that was not captured adequately by prediction stability alone. This observation has implications for the evaluation of post-hoc GNN explainers. Evaluating an explanation method only on instances for which the underlying model is accurate or prediction-stable may leave an additional source of variability unexamined. Contemporary evaluations of GNN explainability increasingly recognize robustness as a separate criterion alongside properties such as correctness, usability, and computational characteristics [2]. The results of the present study support the practical value of explicitly measuring explanation stability under controlled perturbations rather than treating stable predictions as a proxy for stable explanations.

H5 produced a different result and provides an important counterpoint to the explanation-overlap findings. Explanation availability remained above 98% at every masking rate, and the rate of unavailable explanations did not increase monotonically with perturbation severity. The unavailable counts were 12, 16, 17, and 14 at masking rates of 1%, 5%, 10%, and 20%, respectively. H5 was therefore not supported. This result indicates that the primary stability effect was not a simple consequence of GraphLIME increasingly failing to return explanations under stronger perturbation. In most perturbation observations, a complete explanation remained available; what changed substantially was the identity of the features appearing in that explanation.

This distinction between explanation availability and explanation stability is methodologically important. An unavailable explanation represents a failure to obtain a complete top-10 feature set under the predefined criteria, whereas a valid explanation with low Jaccard similarity represents a successfully generated explanation that differs from its baseline counterpart. Conflating these cases by assigning zero similarity to unavailable explanations would artificially combine two different phenomena. By retaining unavailable cases as missing for overlap calculations and analysing availability separately, the present design isolates changes in explanation composition from failures of explanation generation. The high availability rates observed across all four masking conditions strengthen the interpretation that the decline in Jaccard similarity reflects changes among successfully generated explanations rather than an increasing prevalence of failed explanations.

One possible explanation for the observed sensitivity lies in the local nature of GraphLIME itself. GraphLIME constructs feature-wise kernels from the local graph neighbourhood and estimates sparse feature importance through an HSIC-Lasso objective [7]. Masking active feature entries changes the local feature distributions from which these kernels are constructed. Consequently, even when the frozen GCN remains on the same side of its categorical decision boundary, the statistical relationships represented within the local GraphLIME optimization problem can change. The selected sparse feature set may therefore change without requiring a change in the predicted class. This interpretation follows from the structure of the method and is consistent with the experimental observations, but the present experiments do not identify a unique causal mechanism for the instability. Establishing whether changes arise primarily from kernel construction, sparse feature competition, local neighbourhood characteristics, or interactions among these factors would require additional targeted experiments.

The results should also not be interpreted as showing that GraphLIME is generally unreliable. Stability is only one property of an explanation method, and sensitivity to perturbed inputs is not necessarily equivalent to incorrectness. In particular, if a perturbation changes information that the model uses internally while leaving the final argmax class unchanged, a changed explanation may reflect a genuine alteration in the model's local behaviour rather than an error by the explainer. Conversely, an explanation method that remains unchanged under every perturbation would not necessarily be desirable if the model's internal dependence on features has genuinely changed. The present experiment therefore measures empirical explanation stability; it does not establish a ground-truth threshold at which an explanation should be regarded

as correct or incorrect.

This consideration also limits the interpretation of the top-10 Jaccard metric. Jaccard similarity captures membership agreement between selected feature sets, but it does not represent changes in coefficient magnitude or feature ranking within the overlapping set. Two explanations may therefore obtain high Jaccard similarity while assigning substantially different importance coefficients to the same features, or obtain lower similarity because features near the selection boundary exchange positions. Accordingly, the current top-10 analysis provides a deliberately interpretable measure of set-level stability rather than a complete characterization of explanation variation.

Within these boundaries, the study provides evidence that prediction stability alone is insufficient for characterizing the behaviour of local GraphLIME explanations under feature perturbation. The strongest result is not merely that explanation overlap decreased as masking increased, but that this decrease persisted after conditioning on prediction agreement and exhibited little node-level coupling with changes in prediction stability. At the same time, explanation availability remained high and showed no monotonic deterioration. Together, these findings separate three properties that can otherwise be conflated: whether a prediction is preserved, whether an explanation can be generated, and whether the generated explanation remains similar to its baseline form. For the fixed Cora graph, frozen GCN, and GraphLIME configuration studied here, these properties behaved differently under controlled feature masking.

The broader implication is that robustness evaluation of post-hoc GNN explanations benefits from treating explanation behaviour as an object of evaluation in its own right. Recent GNN-explainability literature identifies robustness and stability as continuing challenges and evaluation considerations [2, 3]. The present results contribute a controlled empirical case study of this issue for GraphLIME: stable categorical predictions and high explanation availability can coexist with substantial changes in the selected local explanatory features. Whether the same pattern generalizes across datasets, GNN architectures, explanation methods, perturbation operators, and alternative stability metrics remains an empirical question and motivates the limitations and future research directions considered in Section 6.

Limitations

The findings of this study should be interpreted within the boundaries of its controlled experimental design. The study was designed

to isolate the relationship between feature perturbation, prediction stability, and GraphLIME explanation stability rather than to establish general conclusions about all graph neural networks or graph explanation methods. Several limitations therefore define the scope within which the results should be understood.

First, the experiments were conducted on a **single citation-network dataset, Cora**. Cora provides a well-established setting for node-classification experiments, but its graph structure, sparse node features, class distribution, and neighbourhood characteristics represent only one type of attributed graph. Explanation behaviour may differ on larger graphs, graphs with denser or continuous node attributes, heterophilous networks, heterogeneous graphs, or graphs from other application domains. The observed decrease in GraphLIME explanation overlap should therefore be interpreted as an empirical result for the fixed Cora setting rather than as evidence that the same quantitative behaviour will occur across graph datasets.

Second, the predictive model was restricted to **one two-layer GCN architecture and one validation-selected checkpoint**. Freezing the model was important for the present experimental design because it ensured that changes observed after perturbation could not be attributed to model retraining or parameter variation. However, this control also limits the generality of the findings. Different GNN architectures may encode and aggregate feature information differently, potentially producing different relationships between feature perturbation and explanation stability. Architectures based on attention, alternative message-passing operators, different neighbourhood depths, or other graph-learning mechanisms may therefore exhibit different stability characteristics. Similarly, the present study does not quantify variability across independently trained instances of the same GCN architecture.

Third, the explanation analysis considered **GraphLIME only**. GraphLIME constructs local feature explanations using neighbourhood-based kernels and a sparse HSIC-Lasso objective, and the stability patterns observed here may depend partly on these methodological characteristics. Other GNN explanation methods can target different explanatory objects, including nodes, edges, subgraphs, or continuous feature attributions, and may respond differently to the same perturbations. Consequently, the results should not be generalized from GraphLIME to post-hoc GNN explainers as a whole. A comparative study applying a common perturbation protocol to multiple explanation methods would be required to determine whether the observed separation between prediction and explanation stability is method-specific or more

broadly shared.

Fourth, the perturbation analysis was intentionally restricted to **feature masking**. Only active feature entries within the frozen two-hop explanation neighbourhood were eligible for masking; graph connectivity, model parameters, and all other experimental components remained unchanged. This design provides a controlled test of sensitivity to feature information but does not characterize robustness to structural perturbations such as edge addition, edge deletion, node removal, or neighbourhood modification. It also does not consider alternative feature perturbations such as additive noise, continuous feature displacement, feature replacement, or distribution-aware corruption. Different perturbation operators can represent substantially different changes to the input and may therefore produce different explanation-stability behaviour.

The masking protocol itself introduces an additional scope constraint. Perturbed active entries were set to zero without subsequent feature re-normalization, and masks at different perturbation rates were sampled separately rather than constructed as nested extensions of lower-rate masks. These choices were fixed before the perturbation analysis and applied consistently to all target nodes. They allow each masking condition to represent a separately sampled perturbation at the specified severity, but they also mean that differences between adjacent rates do not correspond to progressively enlarging the same feature mask. The results therefore characterize stability under the particular randomized masking protocol used in this study.

Fifth, explanation stability was primarily quantified using **top-10 Jaccard similarity**. This measure provides a transparent assessment of membership agreement between the baseline and perturbed feature sets, but it discards information about coefficient magnitudes and within-set feature rankings. For example, two explanations containing the same ten features receive a Jaccard similarity of one even if their estimated importance coefficients differ substantially. Conversely, replacement of a feature near the top-10 selection boundary reduces set overlap even when the underlying coefficient values are similar. The current results therefore characterize **set-level feature-selection stability**, not every possible dimension of explanation stability. Rank-based measures, coefficient-distance measures, or stability analyses across multiple values of K could provide complementary perspectives.

Relatedly, the choice of $K = 10$ fixes the resolution at which explanation overlap was evaluated. This value was held constant throughout the study to ensure comparability across target nodes and perturbation conditions, but stability may depend on explanation-set size. Smaller values of K could emphasize changes among the most highly

weighted features, whereas larger values could produce different overlap behaviour by including lower-ranked coefficients. The study did not conduct a post-hoc sensitivity analysis over multiple values of K , because doing so after observing the primary results would alter the scope of the frozen experimental protocol.

Sixth, the statistical analysis is constrained by the **dependence structure of a single graph**. The target node was treated as the statistical unit, and perturbation repetitions were aggregated within node rather than treated as independent observations. Node-level bootstrap confidence intervals and paired node-level comparisons were used accordingly. Nevertheless, target nodes belong to the same Cora graph and their local neighbourhoods may overlap, so independence between target-node observations cannot be assumed strictly. The node-level procedure avoids the substantially stronger and inappropriate assumption that all 3,880 perturbation observations are independent, but it does not eliminate dependence induced by shared graph structure. Inferential quantities should therefore be interpreted as summaries of the fixed experimental graph rather than as population-level guarantees across independently sampled graphs.

Seventh, each node-masking-rate condition was evaluated using **ten perturbation seeds**. Repeated perturbations provide information about variability induced by random masking, but ten realizations constitute a finite sample of the possible masks available for each target node and masking severity. Prediction-stability proportions are consequently discrete at the node level, and explanation-stability estimates may vary if substantially more perturbation realizations are evaluated. Increasing the number of perturbation seeds would permit finer estimation of within-node perturbation variability, although at increased computational cost because each perturbation requires recomputation of both the frozen GNN output and the local GraphLIME explanation.

Eighth, the GraphLIME implementation reproduces the **HSIC-Lasso objective** used by GraphLIME but does not reproduce the original optimization algorithm exactly. The original GraphLIME formulation employs non-negative least-angle regression for solving the sparse optimization problem, whereas the implementation in this study uses a **non-negative proximal-gradient procedure** applied to the same constrained convex objective. Numerical validation was therefore performed explicitly: all 97 retained baseline solutions satisfied the predefined KKT validation criteria, with residuals below the specified tolerance where applicable. These checks support the numerical consistency of the solutions with the implemented objective, but they do not establish algorithm-

mic equivalence between proximal-gradient optimization and the original non-negative LARS procedure. Accordingly, the present results should be described as an objective-faithful GraphLIME implementation rather than an exact reproduction of the original optimizer.

Ninth, explanation availability was determined using predefined computational validity criteria, including convergence of the optimization procedure and the availability of at least ten non-zero coefficients for constructing a complete top-10 explanation. Of the 3,880 perturbation observations, 59 explanations were unavailable under these criteria. Their Jaccard similarities were retained as missing rather than assigned a value of zero. This treatment avoids conflating explanation-generation failure with explanation disagreement, but it also means that overlap statistics are conditional on explanation availability. Because availability exceeded 98% at every masking rate, the amount of missing explanation-overlap data was limited; nevertheless, the stability summaries do not describe the unavailable observations themselves.

Finally, the study does not possess **ground-truth feature explanations** against which GraphLIME outputs can be evaluated. A change in the selected explanatory feature set cannot therefore be classified automatically as either desirable adaptation or erroneous instability. Feature masking may genuinely alter aspects of the frozen GCN's local behaviour without changing its final predicted class, in which case a changed explanation could reflect a meaningful change in the model rather than a failure of the explanation method. Conversely, explanation changes could also arise from sensitivity of the local surrogate construction or sparse feature-selection procedure. The present experiments distinguish and quantify explanation stability, prediction stability, and explanation availability, but they do not establish explanation correctness or causal faithfulness.

These limitations define the appropriate interpretation of the study rather than invalidate its controlled findings. Within a fixed Cora graph, frozen two-layer GCN, fixed GraphLIME configuration, predetermined target-node cohort, and preregistered feature-masking protocol, the experiments provide consistent evidence that top-10 explanation overlap decreases with increasing masking severity and that this decrease persists when the predicted class is preserved. At the same time, explanation availability remains high and does not deteriorate monotonically. Establishing the extent to which these observations generalize will require experiments across additional graphs, model architectures, explanation methods, perturbation operators, explanation metrics, and independently trained models.

Conclusions

This study investigated the stability of local GraphLIME feature explanations for a frozen graph neural network under controlled perturbations of the input node features. The central objective was to determine whether preservation of a GNN prediction is accompanied by preservation of its corresponding local explanation. Using the Cora citation network, a frozen two-layer GCN, a fixed GraphLIME configuration, and predetermined feature-masking conditions, the study evaluated prediction stability, top-10 explanation stability, and explanation availability as distinct empirical properties.

The results showed a pronounced reduction in explanation stability as feature-masking severity increased. Mean top-10 Jaccard similarity between baseline and perturbed GraphLIME explanations decreased from **0.9015 at 1% masking to 0.4541 at 20% masking**, while prediction stability decreased more modestly from **0.9969 to 0.9588**. Importantly, the decline in explanation overlap persisted when the analysis was restricted to perturbation observations that preserved the baseline predicted class: prediction-conditioned mean Jaccard similarity decreased from **0.9014 to 0.4560** across the same masking range. These findings demonstrate that, within the experimental setting considered here, preservation of the categorical prediction did not imply preservation of the selected explanatory feature set.

The node-level analysis further supported this distinction. Changes in prediction stability and explanation stability between the endpoint masking conditions exhibited little descriptive association, with Spearman and Pearson correlations close to zero. Moreover, **86 of 97 endpoint-valid target nodes (88.66%)** retained the same endpoint prediction-stability proportion while exhibiting decreased explanation overlap. Prediction stability therefore provided limited information about the corresponding change in GraphLIME explanation stability within this experiment.

Explanation availability exhibited a different pattern. Complete GraphLIME explanations remained available for more than **98% of perturbation observations at every masking rate**, and unavailability did not increase monotonically with perturbation severity. Consequently, the observed reduction in explanation overlap was not accompanied by a corresponding systematic deterioration in the ability of the explanation procedure to return a complete top-10 feature set. This distinction reinforces the importance of treating **prediction preservation, explanation preservation, and explanation availability as separate properties** when evaluating post-hoc explanations.

The contribution of this study is therefore a controlled empirical characterization of GraphLIME explanation stability rather than a claim that GraphLIME explanations are generally unreliable. Explanation change does not by itself establish explanation incorrectness, particularly because no ground-truth feature explanations were available and a perturbation may alter aspects of local model behaviour without changing the final predicted class. The conclusions are also bounded to the fixed Cora graph, GCN checkpoint, GraphLIME configuration, feature-masking protocol, and top-10 set-based stability measure used in this study.

Future work can determine whether the observed separation between predictive and explanatory stability extends beyond this experimental setting. Relevant directions include evaluating additional graph datasets and GNN architectures, comparing multiple explanation methods, examining structural and alternative feature perturbations, varying explanation-set size, incorporating rank- and coefficient-sensitive stability measures, increasing the number of perturbation repetitions, and studying independently trained model instances. Such extensions would help establish when explanation instability reflects genuine changes in model behaviour and when it instead arises from sensitivity of the explanation procedure itself.

Overall, the findings demonstrate that **stable predictions alone are insufficient to characterize the stability of local GraphLIME explanations**. Under controlled feature masking, the frozen GCN frequently preserved its categorical prediction while the corresponding explanatory feature sets changed substantially. Explicit evaluation of explanation stability therefore provides information about post-hoc GNN explanations that cannot be obtained from predictive stability alone.

Data and Code Availability

The dataset used in this study is the publicly available **Cora citation-network dataset**, accessed through the Planetoid dataset interface in PyTorch Geometric. The experimental code, configuration files, frozen model checkpoint, processed experimental results, statistical-analysis outputs, and figures required to reproduce and inspect the reported analyses are publicly available in the accompanying GitHub repository:

GraphLIME Explanation Stability Study – GitHub repository

The repository documents the complete experimental pipeline, including baseline GCN training and checkpoint selection, target-node selection, GraphLIME explanation generation and numerical vali-

dation, controlled feature perturbation, perturbation-result validation, statistical analysis, hypothesis evaluation, and figure generation. Raw perturbation-level results and the derived node-level and rate-level summaries used in the manuscript are retained in the repository to support reproducibility and independent inspection of the reported findings.

The experimental protocol and principal analysis artifacts were frozen after validation. Subsequent manuscript preparation did not involve retraining the GCN, replacing target nodes, retuning GraphLIME hyperparameters, modifying the perturbation protocol, or altering the statistical analysis in response to the observed results.

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Appendix A Supplementary Experimental Details

This appendix records supplementary implementation and reproducibility details for the experimental procedure described in Section 3. All settings reported here correspond to the frozen configuration used to generate the results in Section 4. They were not modified during subsequent statistical interpretation or manuscript preparation.

A.1 Baseline GCN Configuration

The predictive model was a two-layer graph convolutional network trained on the predefined Cora training split. Model selection was performed exclusively using validation accuracy, and the test split was evaluated only after the selected checkpoint had been restored.

The frozen baseline configuration was:

Parameter	Value
Dataset	Cora
Dataset interface	PyTorch Geometric Planetoid
Feature preprocessing	NormalizeFeatures
Model	Two-layer GCN
Hidden dimension	16
Activation	ReLU
Dropout probability	0.5
Optimizer	Adam
Learning rate	0.01
Weight decay	0.0005
Maximum training epochs	200
Training/model seed	42
Selected checkpoint epoch	125
Training accuracy at selected checkpoint	0.9929
Validation accuracy at selected checkpoint	0.7840
Test accuracy after checkpoint selection	0.8060

The selected model parameters were subsequently frozen. No retraining or fine-tuning was performed during explanation generation or feature perturbation.

A.2 Target-Node Selection and Baseline Explanation Cohort

After restoration of the frozen GCN checkpoint, correctly classified nodes from the predefined test split were identified. A deterministic random permutation with selection seed 42 was used to select 100 correctly classified test nodes without replacement. The selected node set was fixed before GraphLIME explanation validity was assessed and was not modified in response to subsequent experimental outcomes.

Of the 100 predetermined target nodes, 97 admitted valid baseline GraphLIME explanations. Three nodes—2697, 2625, and 2411—were classified as output-degenerate because the local GCN probability vectors did not provide sufficient variation for construction of a non-degenerate GraphLIME output kernel. These nodes were retained in the diagnostic record and were not replaced. The resulting 97 baseline-explainable nodes formed the fixed cohort for the perturbation study.

All 97 retained baseline explanation problems converged with at least 10 non-zero feature coefficients. Numerical validation of the implemented constrained HSIC-Lasso solutions produced 97 of 97 KKT validation passes. The maximum active-set stationarity residual was 6.59×10^{-8} , the maximum inactive-set violation was 0, the minimum inactive-set slack was 2.80×10^{-8} , and the maximum complementarity residual was 3.79×10^{-8} .

A.3 Frozen GraphLIME Configuration

GraphLIME explanations were constructed from the two-hop neighbourhood of each target node. The complete GCN class-probability vectors were used to construct the local output kernel. Gaussian kernels were constructed for locally varying input features and for model outputs, with bandwidth determined from the median positive pairwise Euclidean distance of the corresponding local variable.

The frozen explanation configuration was:

Parameter	Value
Neighbourhood depth	2 hops
Explanation size	Top 10 features
Regularization parameter ()	0.03
Candidate () values used during calibration	0.001, 0.003, 0.01, 0.03, 0.1, 0.3
Calibration cohort	First 10 frozen selected target nodes
Optimizer	Non-negative proximal-gradient
Maximum optimizer iterations	10,000
Relative coefficient-change tolerance	1×10^{-8}
Coefficient zero tolerance	1×10^{-8}

Parameter	Value
KKT validation tolerance	1×10^{-8}
Kernel	Gaussian
Kernel centring	Yes
Frobenius normalization	Yes

The regularization parameter was selected before the perturbation experiments. A candidate was eligible only when optimization converged and at least 10 non-zero coefficients were available for every calibration node. The selected value, $\lambda = \mathbf{0.03}$, was then held fixed for all remaining baseline and perturbed explanations.

The implementation solves the non-negative HSIC-Lasso objective used by GraphLIME through proximal-gradient optimization. It is therefore objective-faithful to the stated convex formulation but does not constitute an exact optimizer-level reproduction of the original non-negative least-angle regression procedure.

A.4 Feature-Masking Protocol

Perturbations were restricted to active feature entries within the same two-hop neighbourhood used for the target node's GraphLIME explanation. An entry was eligible for masking only when its value in the original normalized feature matrix was non-zero.

For a target node with $\mathbf{n_active}$ eligible local entries and masking rate $\mathbf{r}$, the number of masked entries was

$$m(r) = \min\{n_{active}, \max(1, \lfloor r n_{active} + 0.5 \rfloor)\}$$

The frozen perturbation configuration was:

Parameter	Value
Perturbation type	Active feature masking
Masking rates	1%, 5%, 10%, 20%
Perturbation seeds	101-110
Repetitions per node-rate condition	10
Baseline-explainable target nodes	97
Total perturbation observations	3,880
Graph structure modified	No
GCN retrained after perturbation	No
Post-mask feature renormalization	No
Perturbations accumulated across rates	No
Masks nested across rates	No

Every perturbation observation was generated from a fresh copy of the original normalized feature matrix. Entries selected by the mask were set to zero. Masks at different rates were sampled separately rather than constructed by progressively extending a lower-rate mask.

Random mask generation was deterministic for each combination of target node, masking rate, and perturbation seed. The implementation derived the random seed as

$$s_{derived} = (s \times 1,000,003 + v \times 10,007 + i_r \times 101) \bmod (2^{63} - 1)$$

where $\mathbf{s}$ is the perturbation seed, $\mathbf{v}$ is the target-node identifier, and $\mathbf{i}$ is the masking-rate index.

A.5 Statistical Configuration

The target node was treated as the primary statistical unit. The ten perturbation realizations associated with each node-rate condition were treated as repeated measurements within that node rather than as independent observations.

Analysis setting	Frozen value
Primary statistical unit	Target node
Bootstrap resamples	10,000
Bootstrap confidence level	95%
Bootstrap seed	20260917
Bootstrap interval	Percentile
Paired inferential test	Two-sided Wilcoxon signed-rank
Wilcoxon implementation	Tie-corrected normal approximation
Continuity correction	No
Zero paired differences	Excluded from signed-rank calculation
Multiple-comparison adjustment	Holm procedure
Effect-size summary	Matched-pairs rank-biserial correlation
H4 association measures	Pearson and Spearman correlations
H4 correlation significance test	None; correlations treated descriptively

Unavailable perturbed explanations were retained as missing for explanation-overlap calculations and were analysed separately through explanation-availability statistics. They were never assigned a Jaccard similarity of zero.

A.6 Explanation Availability and Diagnostic Outcomes

Across the 3,880 perturbation observations, 3,821 produced valid complete top-10 perturbed explanations and 59 were unavailable. The rate-specific diagnostic outcomes were:

Masking rate	Total observations	Unavailable explanations	Unavailability rate
1%	970	12	1.24%
5%	970	16	1.65%
10%	970	17	1.75%
20%	970	14	1.44%
Total	3,880	59	1.52%

Of the 59 unavailable explanations, 35 contained fewer than the 10 non-zero coefficients required to construct a complete top-10 feature set, while 24 were classified as optimizer non-convergence. These outcomes were retained as diagnostic information rather than converted into explanation-disagreement scores.

A.7 Reproducibility and Research Artifacts

The experimental repository preserves the frozen model checkpoint, configuration records, baseline explanation outputs, perturbation-level observations, validation outputs, statistical summaries, and manuscript figures. The principal analysis artifacts include node-rate summaries, prediction-conditioned summaries, paired rate contrasts, hypothesis summaries, explanation-availability summaries, and endpoint coupling analyses.

The complete research artifact is publicly available in the accompanying repository:

GraphLIME Explanation Stability Study – GitHub repository

The experimental pipeline was frozen after validation and statistical analysis. Manuscript preparation did not involve replacing target nodes, retraining the predictive model, retuning GraphLIME using perturbation outcomes, modifying the masking protocol, or changing the statistical procedures in response to the final results.